

3.2 Self-Attention (Q, K, V)

Scaled dots, softmax over keys, mix values. Multi-head, then a causal mask.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Self-attention is a weighted sum of **values**, with weights from a comparison of **queries** to **keys**. Every token plays all three roles. That is CMU 11-711's calculation, and Lab 3.

First time this method appears. Self-attention is the calculation inside a transformer layer.

What. A weighted sum of **values**, with weights from **queries** compared to **keys**. Every token plays all three roles. **Why.** RNNs mix left-to-right through h_t . Self-attention mixes any pair in one layer, in parallel. **Architecture.** $X \mapsto Q, K, V$ by three matrices. Scores $QK^T / \sqrt{d_k}$, softmax over keys, mix V . Multi-head: several of those in parallel, then W_O . Causal mask for LMs. **How.** Lab 3: one head by hand, then a causal mask. Zero is the wrong mask fill; use $-\infty$. **Formula.**

Attention $(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$. **Tradeoffs.** + Parallel, path length 1, multi-head specialization. – $O(T^2)$; mixing **keys** instead of values is a common bug; no positions until note 3.3.

1. Q, K, V

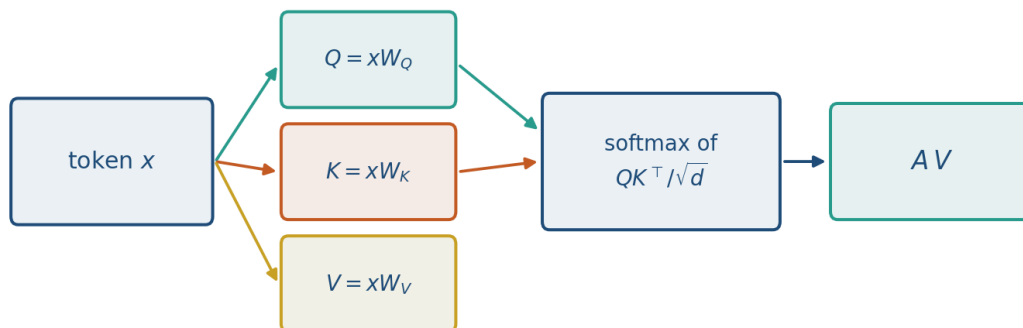
For a sequence matrix $X \in \mathbb{R}^{T \times d}$,

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Scores are scaled dot products, then a softmax over keys:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right), \quad \text{Attention}(Q, K, V) = AV.$$

Query asks, key is asked, value is what you mix in



Query: what this token is looking for. **Key:** what another token looks like to be found. **Value:** what you actually add into the representation.

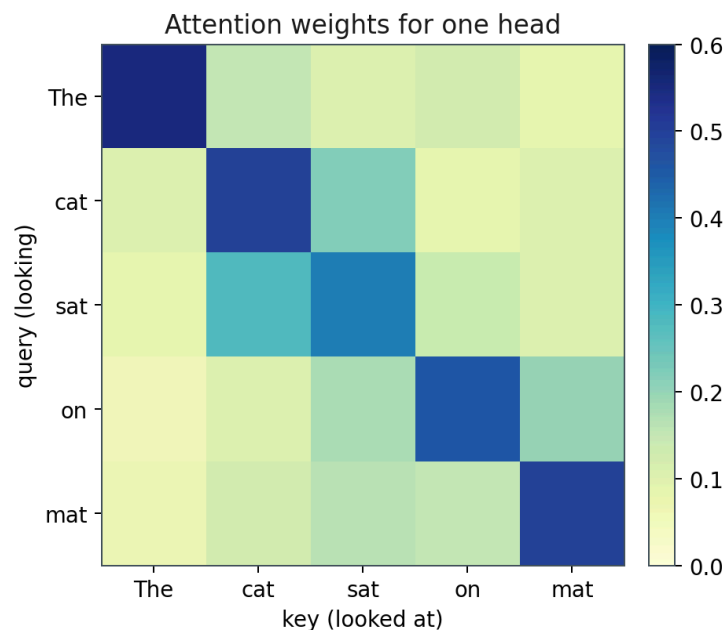
Row t of QK^T is “how much token t wants each position.” Softmax makes that a distribution. Then you mix **values**, not keys. Mixing keys is a common lab bug.

Why $\sqrt{d_k}$. If the coordinates of q and k are roughly independent with variance 1, the dot product has variance d_k . Large d_k pushes softmax toward a one-hot. Dividing by $\sqrt{d_k}$ puts the logits back in a range where softmax still mixes. Skipping the scale, or dividing by d_k instead, is the other common exam slip.

Self versus **cross**. Self-attention: Q, K, V all from the same sequence. Cross-attention (encoder–decoder, later BLIP): queries from one sequence, keys and values from another. Same formula.

2. A row of weights

Row t of A is a distribution over the sequence. Large mass on a name is “this pronoun just looked at that name.”



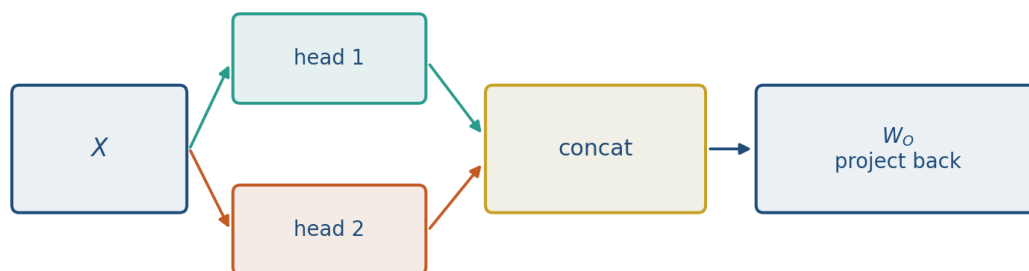
If two keys are equal, the softmax splits mass equally between them (given the same query). That is not a tie-break bug; it is the definition.

3. Multi-head

Several (W_Q, W_K, W_V) in parallel, each on a slice of the channels (head dimension $d_k = d / h$).

Concatenate the head outputs, then a linear W_O . Different heads can track syntax, a name, or a comma.

Multi-head: several small attentions, then one linear map

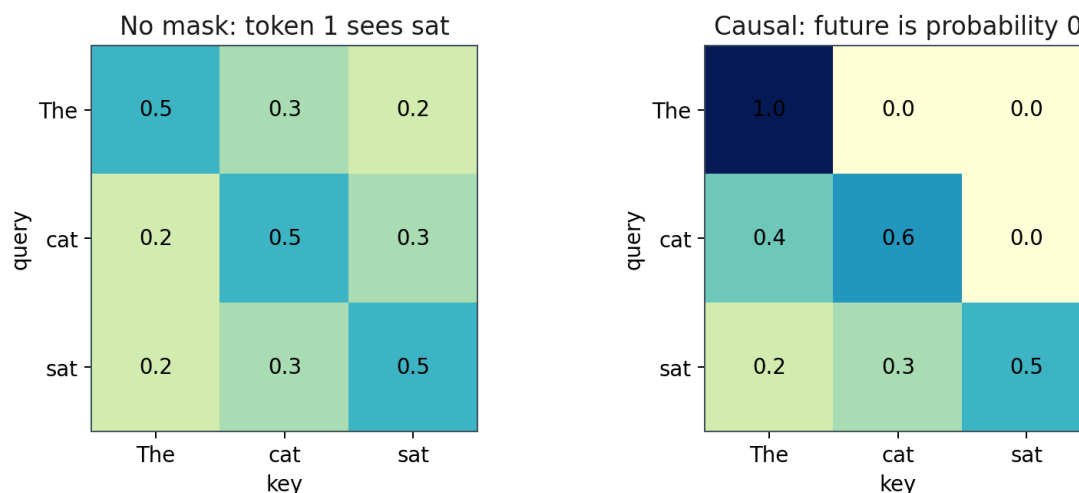


Lab 3 computes **one** head by hand. You do not need eight heads on the board. You need: concat is not a residual, and W_o is part of attention's parameter count in note 3.3.

4. Masks

For next-token prediction you **mask** future keys (set scores to $-\infty$ before softmax). Otherwise the model cheats. BERT does not mask the future; it masks random **tokens** and reconstructs them (note 3.4). Those are two different uses of the word *mask*.

Zero is the wrong fill-in: $\text{softmax}([1, 0])$ still puts mass on the "zero." $-\infty$ becomes probability 0.



5. Teaching this note

~18 minutes. Define Q, K, V in one sentence each, write the scaled-dot formula, grind the 2×2 table (including $\sqrt{d_k}$), then the causal mask. Play **the QKV and softmax stretch** of 3Blue1Brown attention (about **4:00–16:00**). Lab 3 is $T = 4$; do not start that notebook until this arithmetic is done once by hand.

6. Worked example

$T = 2$, $d_k = 2$, skip W_Q and take scores $S = QK^T$ already:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \frac{S}{\sqrt{2}} \approx \begin{bmatrix} 1.41 & 0 \\ 0 & 1.41 \end{bmatrix}.$$

Row 1 softmax: $e^{1.41} \approx 4.10$, $e^0 = 1$, so $[4.10, 1] / 5.10 \approx [0.80, 0.20]$.

Row 2 is the same on the diagonal: $A \approx \begin{bmatrix} 0.80 & 0.20 \\ 0.20 & 0.80 \end{bmatrix}$.

$S = \text{diag}(2, 2)$, $d_k = 2$. Scale, softmax, mix values.

row	scaled scores	softmax A	mix of $V = I$
1	[1.41, 0]	[0.80, 0.20]	[0.80, 0.20]
2	[0, 1.41]	[0.20, 0.80]	[0.20, 0.80]

If $V = I$, then $AV = A$. Each output is 80% "self" and 20% "the other token."

Causal mask on row 1: set the future score to $-\infty$. Then row 1 softmax is $[1, 0]$: token 1 cannot look at token 2.

Without the $\sqrt{2}$, row 1 would be softmax($[2, 0]$) $\approx [0.88, 0.12]$: more peaked. That is the scale's job.

7. Where students get stuck

- Softmax over the wrong axis (columns instead of keys / last dim).
 - Dividing by d_k instead of $\sqrt{d_k}$, or skipping the scale.
 - Masking with 0 instead of $-\infty$.
 - Mixing keys instead of values.
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8. Video

Watch [3Blue1Brown: Attention in transformers, visually explained](#).

Pause when queries and keys meet as a grid of dots, when softmax turns a row into a distribution, and when values are mixed. That grid is Lab 3's heatmap.

The matching university lecture is CMU 11-711 (Neubig): scaled dot-product, multi-head, then the causal mask for LM training. We do not copy those slides.

9. Practice

1. If two tokens have identical keys, what does the softmax do to their weights?
2. Why divide by $\sqrt{d_k}$?
3. Write A for $T = 2$ with $Q = K = V = I$ (identity), no mask. What is AV ?
4. Softmax the row $[0, 2]$. Then softmax $[0, 2] / \sqrt{2}$. Did the larger entry gain or lose probability after scaling?
5. Scores $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$, causal mask (upper triangle $-\infty$). Write the masked score matrix, then A (row-wise softmax).
6. Self-attention versus cross-attention: which matrices come from which sequence?
7. Two heads each emit a 2-D vector. After concat, what width does W_O map back to if the model width is $d = 4$?